Skills Summary

Geometric Series, p-Series, and the nth-Term Test

Use this reference while completing your worksheet or when studying.

Skill 1 — Geometric series

Test: consecutive terms have a constant ratio r. **Verdict:** converges exactly when $|r| < 1$. **Value:** $\dfrac{\text{first term actually written}}{1 - r}$. The starting index matters only through that first term — $\sum_{n=0}$ and $\sum_{n=1}$ of the same expression have different first terms and therefore different sums.

Example: Find $\displaystyle\sum_{n=1}^{\infty} 3\left(\frac{1}{4}\right)^n$.

Step 1. Consecutive terms differ by the constant factor one quarter, so this is geometric and the closed form applies; the only care needed is which term is first.

Step 2. The sum starts at $n = 1$, so the first term is $3 \cdot \frac{1}{4} = \frac{3}{4}$ and $r = \frac{1}{4}$, giving $\dfrac{3/4}{1 - 1/4}$.

$\Rightarrow$ **1**

Try it: Find $\displaystyle\sum_{n=1}^{\infty} 5\left(\frac{1}{3}\right)^n$.

check: $\frac{5}{2}$

Skill 2 — p-series

Test: the terms are $\dfrac{1}{n^p}$ for a constant p. **Verdict:** converges exactly when $p > 1$; diverges when $p \leq 1$. The boundary case $p = 1$ is the harmonic series, which diverges even though its terms go to zero — the standing reminder that shrinking terms are not enough. Values worth knowing: $p = 2$ gives $\frac{\pi^2}{6}$, $p = 4$ gives $\frac{\pi^4}{90}$.

Example: Does $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^6}$ converge, and to what?

Step 1. The terms are one over a fixed power of n, so this is a p-series and the exponent alone decides convergence.

Step 2. Here $p = 6 > 1$, so it converges; its value is the standard even-exponent result $\frac{\pi^6}{945}$.

$\Rightarrow$ $\boldsymbol{\frac{\pi^6}{945}}$

Try it: Does $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2}$ converge, and to what?

check: $\frac{\pi^2}{6}$

Skill 3 — The nth-term test

One direction only. If $\lim_{n\to\infty} a_n \neq 0$, then $\sum a_n$ diverges. If the limit *is* zero, the test says nothing — the series may converge or diverge, and the harmonic series shows both outcomes are possible. So use it as a cheap first screen: take the limit, and if it is not zero you are finished.

Example: What does the nth-term test say about $\displaystyle\sum_{n=1}^{\infty} \frac{4n+3}{2n+1}$?

Step 1. Before reaching for any comparison, check the cheapest thing: whether the individual terms even shrink to zero.

Step 2. Dividing top and bottom by n gives $\dfrac{4+3/n}{2+1/n}$, whose limit is $\dfrac{4}{2}$ — not zero, so the series diverges.

$\Rightarrow$ **2**, so the series diverges

Try it: Find $\displaystyle\lim_{n\to\infty} \frac{7n-1}{3n+5}$ and say what it tells you about the series.

check: $\mathbf{\frac{7}{3}}$, so it diverges